

Foundations of Statistics for in ML & AI

Target outcome:

Participants stop treating statistics as magic and start seeing it as structured common sense.



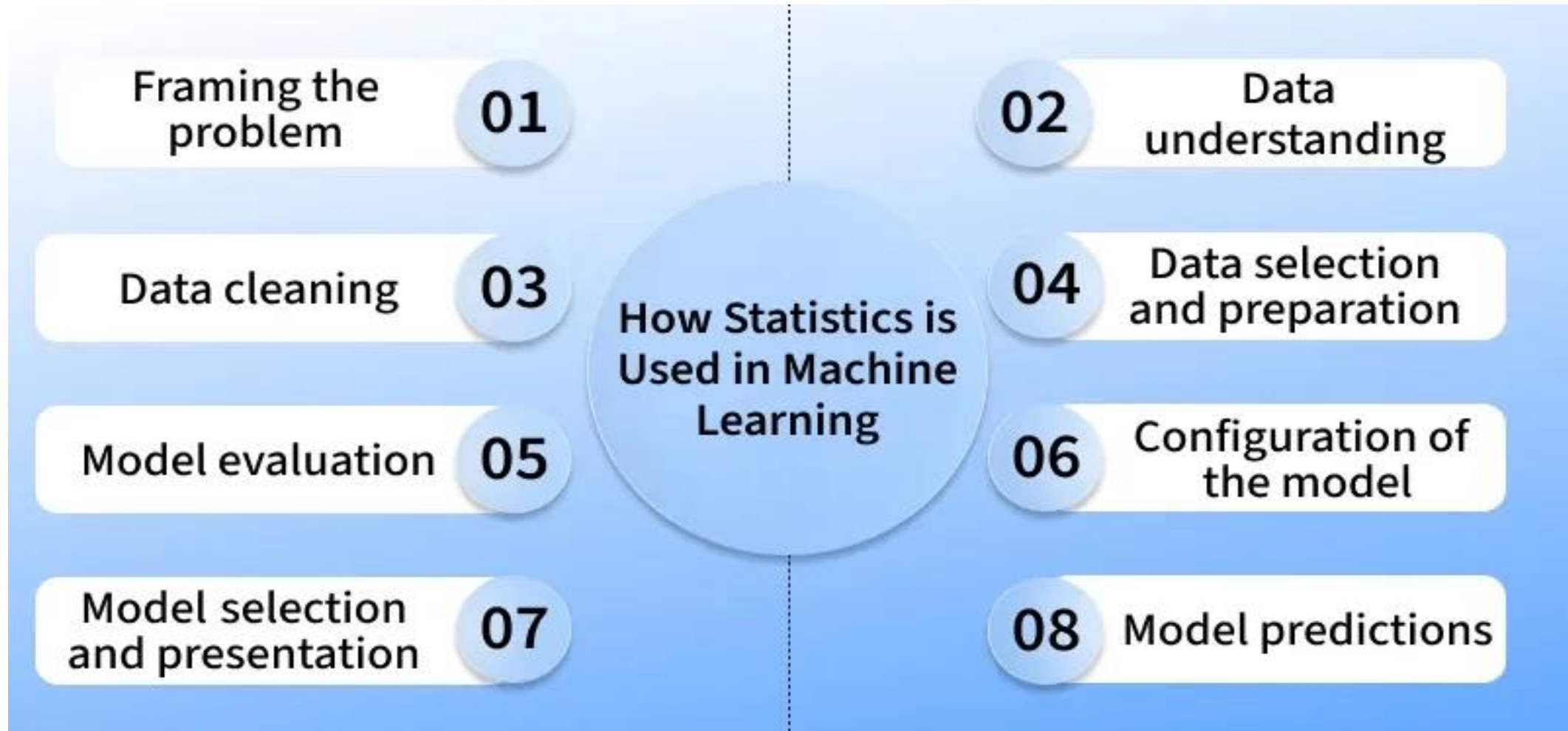
Why statistics matters in AI and ML

Statistics is the science of extracting insight from data. It organises, analyses and interprets information to uncover patterns and make decisions under uncertainty.

Key messages

- ML models are opinions expressed in math: modern ML is built on top of statistical methods
- Statistics is how we measure uncertainty and risk
- Most AI failures are statistical misunderstandings, not model failures
- It helps in understanding data distribution, variability and selecting the most useful features.
- It is used to validate model results and make decisions under uncertainty using hypothesis tests, confidence intervals and Bayesian methods.

Why statistics matters in AI and ML



Descriptive statistics: Understanding the basics

Before training models, we perform exploratory data analysis (EDA), a process that relies on descriptive statistics to summarise key characteristics of the data.

These summaries inform us about;

- the central tendency and variability of each feature,
- helping identify outliers,
- data quality issues, and
- preprocessing needs.

Understanding these properties is a prerequisite for building effective models and choosing appropriate machine learning algorithms.



Descriptive Statistics

Mean (The Sensitive Average)	Median (The Stubborn Middle)	Mode (The Crowd Favourite)
• Simple Summary: The balance point of the data where every value has a say. It is the arithmetic average of values. Common in measuring centrality and in loss functions like mean squared error (MSE). • How/Where Used: Used when data is symmetric and "normal" (no crazy spikes). • Example: Calculating the average height of students in a classroom. If customer purchase values are increasing, the mean detects shifts in behaviour. • Practitioner Insight: In ML, we use Mean to normalise features so they all sit on a similar scale.	• Simple Summary: The literal middle value when data is sorted. It doesn't care about the size of the numbers at the ends; it only cares about their position. More robust to outliers than the mean. • How/Where Used: Used when data is "skewed" or has extreme outliers. • Example: Reporting "Median Household Income" (because one billionaire would ruin the mean). In income data, the median better reflects a "typical" case in the presence of skewed wealth. • Practitioner Insight: Use the median to fill in missing data (imputation) when your dataset has outliers you don't want to delete.	• Simple Summary: The most frequent value or category i.e. the most frequently occurring value or majority voting (as in some ensemble methods). • How/Where Used: Used for categorical data where math doesn't apply. • Example: Determining the most popular t-shirt size (Small, Medium, or Large) in an inventory. Finding the most common browser used by site visitors.

Descriptive Statistics

```
import numpy as np
from scipy import stats

# Dataset A no outlier
data_clean = np.array([10, 12, 11, 13, 12, 14, 15, 16, 12, 17])

# Dataset B with outlier
data_outlier = np.array([10, 12, 11, 13, 12, 14, 15, 16, 12, 100])

def calculate_statistics(dataset, label):

    mean_val = np.mean(dataset)
    median_val = np.median(dataset)
    mode_val = stats.mode(dataset, keepdims=True).mode[0]

    print(f"\nStatistics for {label}")
    print("Mean:", round(mean_val, 2))
    print("Median:", median_val)
    print("Mode:", mode_val)

calculate_statistics(data_clean, "Dataset WITH Outlier")
calculate_statistics(data_outlier, "Dataset WITHOUT Outlier")
```

...

Statistics for Dataset WITH Outlier

Mean: 21.5

Median: 12.5

Mode: 12

Statistics for Dataset WITHOUT Outlier

Mean: 13.2

Median: 12.5

Mode: 12

Descriptive Statistics



Variance & Standard Deviation (The "Nervous Energy")

If the **Mean** is the *anchor* then **Variance and Standard Deviation** describe how much the data is trying to pull away from that anchor.

Variance: The "Raw Math Engine"	Standard Deviation: The "Back to Reality" Ruler
It's the squared of the average of how far every point is from the center, i.e., the mean of squared deviation from the mean	Measures how spread out the values are from the mean. The square root of Variance. It brings back into the original units we understand (dollars, centimeters, etc.)
Why we square it: If we didn't square the distances, the negative numbers (points below average) and positive numbers (points above average) would cancel each other out to zero. Squaring makes everything positive and "punishes" outliers—making big gaps look even bigger.	A low SD implies that the data is "calm", data points are clustered near (hugging) the mean, while a high SD indicates the data is "frantic", greater variability. The data points are scattered everywhere (High risk/unpredictability).
The Catch: Because we squared everything, the units are weird (e.g., "squared dollars" or "squared centimeters"). It's hard to visualize.	



Descriptive Statistics

```
import numpy as np

# Dataset A no outlier
data_clean = np.array([10, 12, 11, 13, 12, 14, 15, 16, 12, 17])

# Dataset B with outlier
data_outlier = np.array([10, 12, 11, 13, 12, 14, 15, 16, 12, 100])
```

```
def calculate_stats(data, label):
    mean_val = np.mean(data)
    median_val = np.median(data)
    std_val = np.std(data)

    print(f"\n{label}")
    print("Mean:", round(mean_val, 2))
    print("Median:", median_val)
    print("Standard Deviation:", round(std_val, 2))
```

```
calculate_stats(data_clean, "Dataset A No Outlier")
calculate_stats(data_outlier, "Dataset B With Outlier")
```

...

Dataset A No Outlier
Mean: 13.2
Median: 12.5
Standard Deviation: 2.14

Dataset B With Outlier
Mean: 21.5
Median: 12.5
Standard Deviation: 26.22

Dataset A No Outlier

Mean around center

Median around center

Standard deviation small

Interpretation

Data is stable

Typical value is reliable

Dataset B With Outlier

Mean jumps upward a lot

Median barely moves

Standard deviation explodes

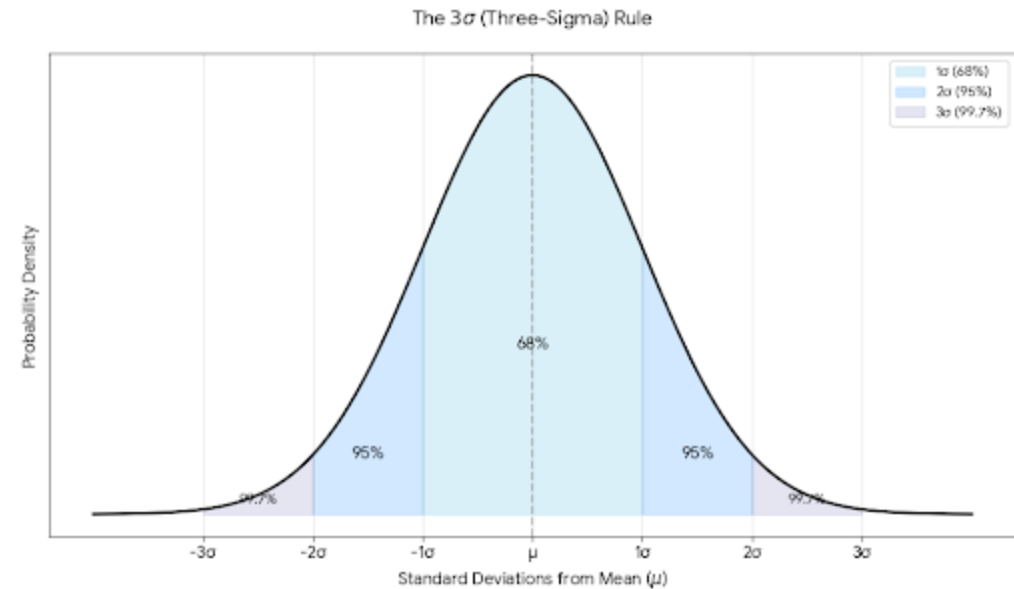
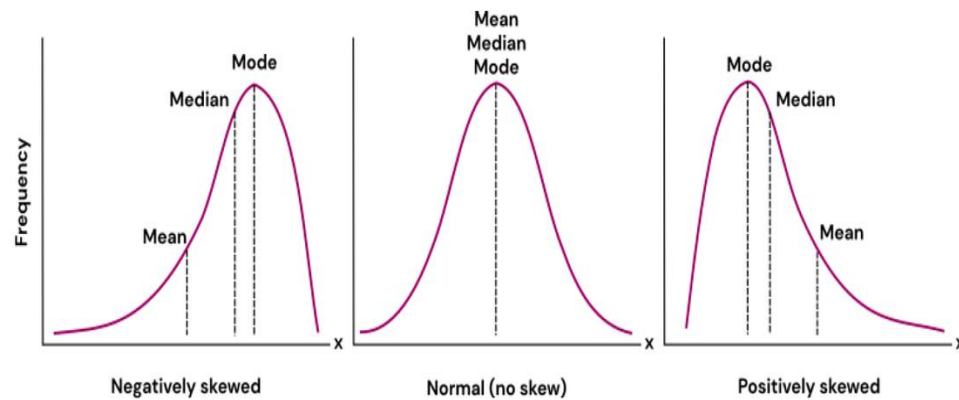
Interpretation

Mean got emotionally hijacked

Median stayed calm

Standard deviation started sweating

Descriptive Statistics



Probabilities: The language of uncertainty

- Modelling by using **machine learning** exists **because of uncertainty**.
 - If we could perfectly map inputs to outputs, there would be no need for models. But real-world data is messy, incomplete and noisy—so we model **likelihoods** instead of certainties.
- Learning about probabilities lays the fundamentals of everything machine learning and artificial intelligence (AI). Theories in probabilities allow us to understand the data we used to model in a beautiful and elegant way.
- It plays a critical role in modelling uncertainties in ML models predictions, and helps us quantify likelihood, probability and certainties for a statistical model so we can confidently measure the outcome models we create.

Basics Terminologies

- **Random variable:** A numerical representation of an outcome of a random phenomenon. It's a variable whose possible values are numerical outcomes of a random process.
 - **Discrete random variable:** A random variable that can take on a finite or countably infinite number of distinct values. For example, the outcome of a coin flip (Heads = 1, Tails = 0), or the number of spam emails received in an hour.
 - **Continuous random variable:** A random variable that can take on any value within a given range. For example, the height of a person, the temperature in a room or the amount of rainfall.
- **Event:** A set of one or more outcomes from a random process. For example, rolling an even number on a die (outcomes: 2, 4, 6) or a customer churning.
- **Outcome:** A single possible result of a random experiment. For example, flipping a coin yields either "Heads" or "Tails."

Probabilities: Basic Terminologies

- **Probability $P(A)$:** A numerical measure of the likelihood that an event A will occur, ranging from 0 (impossible) to 1 (certain).
- **Conditional probability $P(A|B)$:** The probability of event A occurring, given that event A has already occurred. This step is crucial in ML, as we often want to predict an outcome given specific features.
- Probability is a measure of how likely an event is to happen, from 0 (impossible) to 1 (certain). So, in machine learning, it often takes the form of conditional probability
- **Example:** A logistic regression model might say
 - Given age = 45, income = N80,000, and prior history of churn,
 - the probability of churn is 0.82.”
- This example doesn't mean that the customer will churn—it's a belief based on the statistical patterns in the training data.

Distributions: Modelling how data behave

- A **probability distribution** is a mathematical function that describes the possible values and likelihoods that a random variable can take within a particular range.
- Understanding distributions is crucial in ML because data rarely exists as single, isolated points; it has a structure and a shape.
 - **Discrete distribution:** Applies to variables that take on distinct, countable values (for example, coin flips, word counts).
 - **Continuous distribution:** Applies to variables that can take any value within a range (for example, height, weight, time).
- Making the right assumptions about your data's distribution is critical—many machine learning algorithms rely on these assumptions for both model selection and interpretation.
- Incorrect assumptions can lead to biased estimates, misaligned loss functions and ultimately, poor generalization or invalid conclusions in real-world applications.

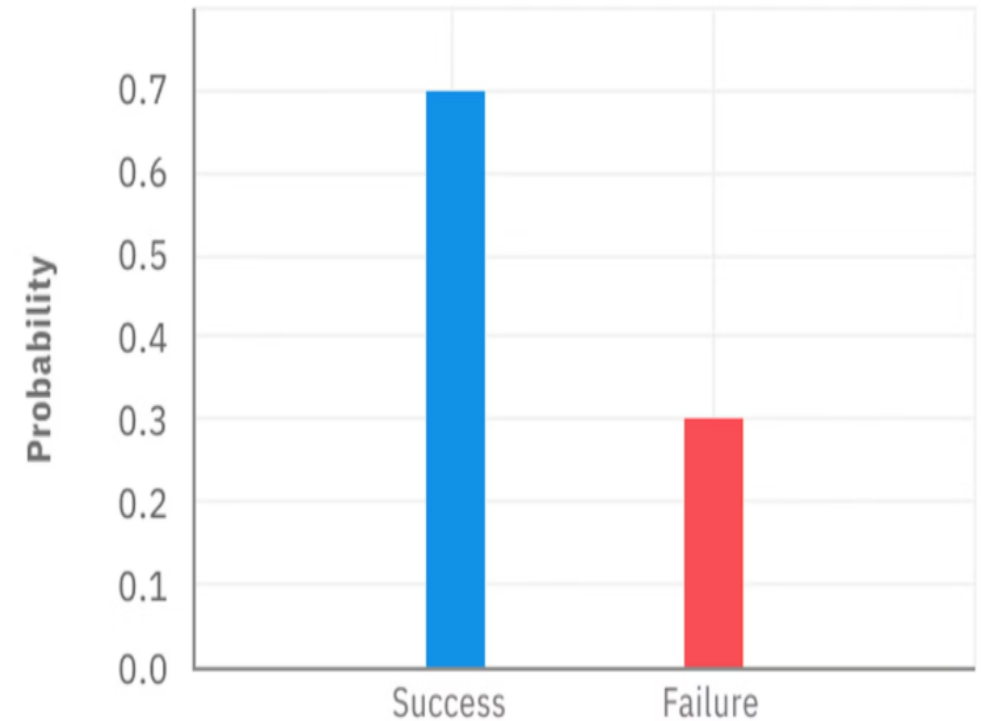
Distributions: Modelling how data behave

Probability distributions underpin:

- **Error modelling:** Assumptions about residuals in regression (often Gaussian).
- **Loss functions:** MSE corresponds to Gaussian assumptions; cross-entropy to Bernoulli or logistic.
- **Model design:** Classification targets are often modelled through Bernoulli; latent variables in deep generative models use Gaussian priors.
- **Generative AI:** Sampling from learned high-dimensional distributions is fundamental to models like generative adversarial networks (GAN) and VAEs.

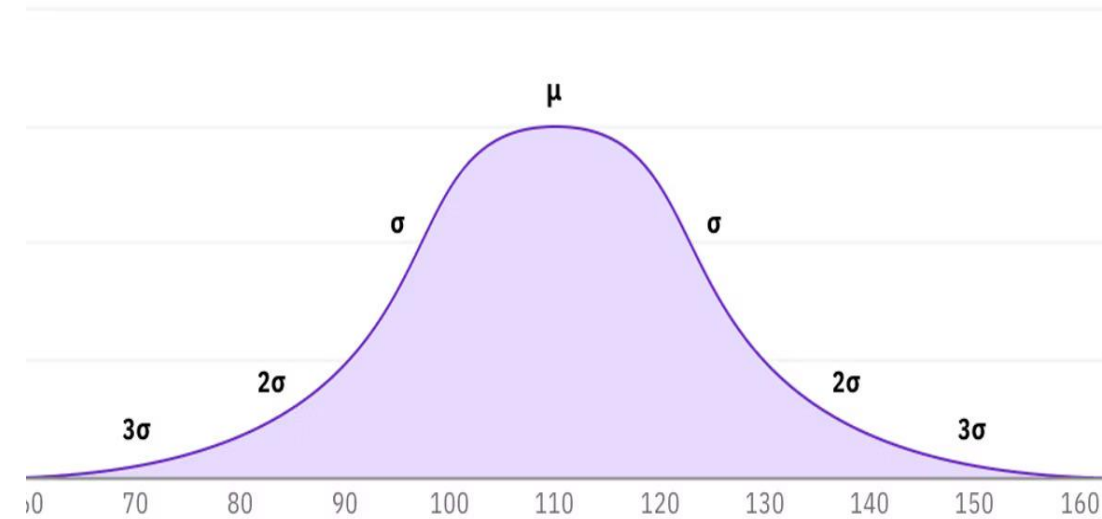
Example of discrete distribution: Bernoulli trials

- The Bernoulli distribution models the probability of success or failure in a single trial of a discrete random event.
- That is, it only has two outcomes: 1 (success) or 0 (failure). It's the simplest type of distribution used in statistics, yet it forms the foundation of many classification problems in machine learning.
- For example, if you were to flip a coin 10 times, and you get 7 heads (success) and 3 tails (failure), the probability mass function (PMF) can be graphed as:



Example of continuous distribution: Gaussian (normal) distribution

- The normal distribution describes a continuous random variable whose values tend to cluster around a central mean, with symmetric variability in both directions.
- It's ubiquitous in statistics because many natural phenomena (height, test scores, measurement errors) follow this pattern, especially when aggregated across samples.
- Imagine you record the heights of 1,000 adults. Plotting this data reveals a bell-shaped curve: most people are close to the average, with fewer at the extremes. This shape is captured by the probability density function (PDF) of the normal distribution.



Hypothesis Testing

Hypothesis testing is a method that compares two opposite assumptions about a population and uses data from a sample to determine which assumption is more likely to be true.

- **Null (H_0) vs. Alternative (H_a) :**
 - H_0 assumes there is no effect or relationship in the population.
 - H_a suggests there's an effect or relationship in the population.
- **Type I and Type II Errors:**
 - **Type I** error is rejecting a true null hypothesis
 - **Type II** is failing to reject a false null hypothesis.
- **p-values:** Measure the probability of obtaining the observed results under the null hypothesis.
 - The "Surprise Meter." If the H_0 is true, how weird is our data?
 - *Small p-value* (< 0.05): "This is too weird to be a coincidence. Reject H_0 ."
 - *Large p-value* (> 0.05): "There's a 50% chance this size difference happened by random luck. Keep H_0 ."

Confidence Intervals

- **Estimation** is the process of making inference on unknown population parameters from sample data.
 - **Point estimate** is a single value that represents our best approximate value for an unknown population parameter.
 - **Interval estimate** is a range of values that is likely to contain the true value of a population parameter.
- **A confidence interval** is an estimate of the range in which the true population mean is likely to lie. It is based on the sample mean and standard deviation and provides a measure of the uncertainty in the estimate.
 - **Wide interval:** " The estimate is too **uncertain**, so it could be anything."
 - **Narrow interval:** "The estimate is very **reliable**."

The Testing Toolbox

Test	When to Use it	Example
t-Test	Small sample, unknown variance.	Comparing the test scores of Class A vs. Class B.
z-Test	Large sample ($n > 30$), known variance	Testing if the average height of a whole country is exactly 175cm.
ANOVA	Comparing 3+ groups .	Do three different diets result in different weight loss amounts?
Chi-Square	Categorical data (Labels).	Is there a link between "Gender" and "Preference for Sci-Fi movies"?

- **t-Tests and z-Tests:** Compare means to assess statistical significance.
- **ANOVA (Analysis of Variance):** Compares means across multiple groups to determine if they differ significantly.
- **Chi-Square Tests:** Assess the association between categorical variables.

Covariance vs Correlation & Correlation vs. Causation

Covariance vs Correlation in ML

- **Covariance:**
 - Measures whether two variables move together.
 - Positive means they increase together. Negative means one increases while the other decreases.
 - Covariance tells you *direction of relationship*, but the scale depends on units, so the number itself is hard to interpret.
- **Correlation:**
- Is a standardized version of covariance.
- It measures both **direction and strength**, always between minus one and one.
- Correlation lets you compare relationships across features and decide which variables are more strongly related to a target or to each other.

Correlation vs Causation in ML

- Correlation does not imply causation.
- Models learn correlations, not causes.
- A feature can improve predictions without being the true driver.
- Risk: Using correlated but non causal features can break models when data distributions change.

